

Zen Nguyen

(672) 673-2613 | ttn54@sfu.ca | linkedin.com/in/zennguyen1305 | github.com/ttn54 | zennguyen.me

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, HTML/CSS
- **Backend & Systems:** Node.js, Express, RESTful APIs, Docker, Git
- **AI & Data:** RAG Architecture, Vector Databases (FAISS/ChromaDB), LangChain, OpenAI API, Anthropic API

TECHNICAL PROJECTS

SFU-Course-Tracker

Nov 2025 - Jan 2026

Personal Project · TypeScript, React, Vite, CSS3, FastAPI, Python 3.12, SQLite, SQLAlchemy ORM, JWT Authentication, Docker, GitHub Actions

- Developed a full-stack web application for SFU students to search, filter, and track course availability, enabling users to monitor course availability with notifications.
- Engineered a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting, increasing data throughput by 10x while respecting server constraints, demonstrating proficiency in building **highly-available applications**.
- Designed and implemented a custom parser to tokenize and evaluate nested boolean prerequisite strings, converting unstructured text into a deterministic Abstract Syntax Tree (AST), showcasing complex logic implementation.
- Migrated full-stack platform from Azure to AWS (EC2), implementing **Docker** container orchestration to optimize deployments and reduce infrastructure costs, aligning with **CI/CD pipelines** and cloud platform experience.

TitanSwarm

Jan 2026 - Present

Personal Project · Python, LangChain, FAISS, Streamlit, SQLite, Gemini API, JobSpy, Pytest

- Architected an autonomous AI Co-Pilot in Python that automates end-to-end job application workflows, enabling the discovery, extraction, and parsing of real-time Software Engineering postings.
- Built a zero-hallucination RAG (Retrieval-Augmented Generation) engine using LangChain, Gemini APIs, and a sandboxed FAISS Vector Store, delivering uniquely tailored, ATS-optimized PDF resumes for each job using only verified user data.

TitanStore

Jan 2026 - Present

Personal Project · Go, Raft Consensus Algorithm, TCP Sockets

- Developed a distributed key-value database in Go from scratch, implementing the Raft consensus protocol for leader election and fault-tolerant log replication across a cluster of nodes.
- Engineered a crash-safe Write-Ahead Log (WAL) with atomic snapshots to guarantee data durability and enable node recovery after failures without data loss.

EDUCATION

Bachelor of Science, Computing Science

May 2025 - Present

Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 - Apr 2025